

Chapters

Chapter 1

Introduction to logic (name of the section)

Goals

Section 1.1

- Recognize that natural language can be **ambiguous**
- Introduce notation that is not **ambiguous**
 - Including logical connectives to build compound

Section 1.2

- Translate between words and symbols

Section 1.3

- Logical equivalences
 - Different symbols can have the same meaning

Chapter 2

Sections 2.1 - 2.5

Sets, functions, sequences, sums & cardinalities

Section 3.2

Growth of functions (big-O and related)

Chapter 5

Sections 5.1 & 5.3

Induction & recursion

Sections 8.1 & 8.2

Solving linear recurrence equations

Chapter 9

Sections 9.1 & 9.3 & 9.5

Relations

Chapter 10

Sections 11.1 & 11.2

Graph theory

Chapter 11

Trees

1.1 Introduction to propositional logic

- Declarative is in the grammar sense, as opposed to:
 - Imperative
 - interrogative
 - exclamatory

- etc

A proposition is a declarative statement that is either true or false (but not both)

Statement is in a broad sense. A statement needs a verb, but it might be a math verb, like '=' or '>'

Examples

- Your math 1019 instructor is the King of Norway (**False** statement)
- Our class is in LSB 105 (**True** statement)
- The population of Toronto at exactly 9am on September 11th 2026, was even, (This might be **true** or **false** but exactly one of those holds)
 - The above is a **proposition**
- "Burlington has an orange psychic aura."
 - Neither true, nor false since it is meaningless drivel
- Hand in Reading Check 0 by next wednesday
 - not **declarative**, it is **imperative**
- Are you in this class?
 - Not **declarative**, it is **interogative**

$$9 + x = 12$$

Truth, depends on x

- wait for predicative function in 1.4